

# Introduction to Cryptography

Ciphers • Number Theory • Hands-on  
Exercises

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# What is Cryptography?

- The science of securing communication
- Core goals: Confidentiality, Integrity, Authenticity
- From simple substitutions → modern ZK proofs
- Symmetric and asymmetric

# Historical Context

- Caesar Cipher (Shift by fixed number)
- Substitution Cipher (unique letter mapping)
- Vigenère Cipher (keyword shifts)

# Caesar Cipher Example

- Plaintext: HELLO
- Shift: 3  $\rightarrow$  KHOOR
- Weakness: easy brute-force (only 25 keys)

# Substitution Cipher

- Arbitrary mapping of letters ( $A \rightarrow Q$ ,  $B \rightarrow R$ , etc.)
- Stronger but breakable with frequency analysis

# Vigenère Cipher

- Uses keyword (e.g. CRYPTO)
- More complex shifts
- Considered unbreakable until advanced analysis

# Exercise Instructions

- Let's pick a simple cipher - Caesar or Vigenère
- We will encode and decode short messages
- Use modular arithmetic for shifts

# Caesar Cipher Exercise

- Encrypt “crypto” with shift = 5
- Decrypt “MJQQT” with shift = 5
- Show your answers!

# Vigenère Cipher Exercise

- Keyword: MATH
- Encrypt “PROOF”
- Decrypt a given message

# From Ciphers to Modern Cryptography

- Shift from puzzles → mathematics
- Built on number theory (primes, modular arithmetic, discrete log)

# Why Number Theory?

- Foundation of RSA, Diffie-Hellman, ECC
- Deals with primes, divisibility, modular arithmetic

# Prime Numbers

- Definition: divisible only by 1 and itself
- Examples: 2, 3, 5, 7, 11, 13...
- Importance: building blocks of cryptography

# Greatest Common Divisor (GCD)

- $\text{GCD}(a, b)$ : largest integer dividing both
- Example:  $\text{GCD}(30, 21) = 3$
- Used in key generation and coprimality checks

# Modular Arithmetic

- "Clock arithmetic"
- Example:  $17 \equiv 5 \pmod{12}$
- Key operations: addition, multiplication, exponentiation mod  $n$

# Why Modulo Matters

- Keeps numbers within finite range
- Core to RSA, ECC, hashing

# Quick Demo

- Compute:  $7 \times 8 \bmod 10 = ?$
- Compute:  $3^4 \bmod 5 = ?$

# Group Discussion

- Which cipher seem stronger?
- How did number theory help in encoding/decoding?
- Where do you see weaknesses?

# Wrap-up

- Cryptography started with ciphers → now uses advanced maths
- Primes, modular arithmetic = core building blocks
- Next steps: modern cryptographic protocols

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